Which Regression Line Fits?

AP Statistics · Unit 5 · Topic 5.5 · TEACHER KEY

CED TETHER

- **Skill 2.C** — Calculate summary statistics, relative positions of points within a distribution, correlation, and predicted response.
- **Skill 4.B** — Interpret statistical calculations and findings to assign meaning or assess a claim.
- **EU DAT-1** — Regression models may allow us to predict responses to changes in an explanatory variable.
- **LO DAT-1.G** — Estimate parameters for the least-squares regression line model. [Skill 2.C]
- **EK DAT-1.G.1** — The least-squares regression model minimizes the sum of the squares of the residuals and contains the point ($\bar{x}$, $\bar{y}$).
- **EK DAT-1.G.2** — The slope, b, of the regression line can be calculated as b = r(s_y / s_x), where r is the correlation between x and y, s_y is the sample standard deviation of the response variable, and s_x is the sample standard deviation of the explanatory variable.
- **EK DAT-1.G.3** — Sometimes, the y-intercept of the line does not have a logical interpretation in context.
- **EK DAT-1.G.4** — In simple linear regression, r^2 is the square of the correlation, r. It is also called the coefficient of determination. r^2 is the proportion of variation in the response variable that is explained by the explanatory variable in the model.
- **LO DAT-1.H** — Interpret coefficients for the least-squares regression line model. [Skill 4.B]
- **EK DAT-1.H.1** — The coefficients of the least-squares regression model are the estimated slope and y-intercept.
- **EK DAT-1.H.2** — The slope is the amount that the predicted y-value changes for every unit increase in x.
- **EK DAT-1.H.3** — The y-intercept value is the predicted value of the response variable when the explanatory variable is equal to 0. The formula for the y-intercept, a, is a = $\bar{y}$ - b$\bar{x}$.

Essential question: How do several regression properties distinguish lines that share the same mean point?

DOK-3 skill: 4.B · **FRQ pattern:** `shared-mean-point-which-line-is-lsrl`

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Keeps Quinn's line using the correct SD ratio and slope
- **reasoning-2** (required) — Explains why the shared mean point is insufficient to choose the LSRL
- **reasoning-3** (required) — Keeps Rowan's 64 percent and interprets response variation in context

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Reversing the standard-deviation ratio
- Treating the mean point as sufficient
- Using r as explained variation

[STAR] THE PROBLEM — annotated

Suppose a solar-farm analyst uses sunshine hours x to predict daily generation y in megawatt-hours (MWh). The table summarizes 12 observed days.

Quinn: "Using $b = r(s_y/s_x)$ gives $\hat{y} = 5.6 + 3.2x$. It passes through $(7, 28)$, so it must be the LSRL. It explains 80 percent of the variation."

Rowan: "Using $b = r(s_x/s_y)$ gives $\hat{y} = 26.6 + 0.2x$. It also passes through $(7, 28)$, so it is equally good. Explained variation is 64 percent."

Solar-farm summary (12 observed days)

x range	$\bar{x}$	$\bar{y}$	s_x	s_y	r
4–10 h	7 h	28 MWh	1.5 h	6 MWh	0.80

FIRST TAKE — one sentence before you work the parts

The two lines agree at 7 sunshine hours. Does that make them equally useful at other times? Give your first thought in one sentence.
Key: Not graded. Credit a committed clause-level comparison supported by one property.

[BOOK] MAKE THE REGRESSION PROPERTIES AGREE (keep on the page)

The least-squares regression line (LSRL) minimizes the sum of squared residuals and contains $(\bar{x}, \bar{y})$. Its slope is $b = r(s_y/s_x)$, its intercept is $a = \bar{y} - b\bar{x}$, and r^2 is the proportion of response variation explained by the linear model. Interpret slope as a predicted response change per unit of x. The intercept predicts at $x = 0$, which may be outside the observed range and unsupported in context.

DOK 1 **(a)** State the mean point that every least-squares regression line must contain.

Key: The mean point is $(\bar{x}, \bar{y}) = (7, 28)$: 7 sunshine hours and 28 MWh.

DOK 2 **(b)** Calculate the regression slope and intercept, and r^2. Give the slope's units.

Key: $b = 0.80(6/1.5) = 3.2$ MWh per hour; $a = 28 - 3.2(7) = 5.6$ MWh; $r^2 = (0.80)^2 = 0.64$.

DOK 3 **(c)** Which parts of Quinn's and Rowan's claims should a report keep? Use the slope formula to choose a line, explain why their shared mean point does not make both lines correct, and interpret the correct percent of variation explained. Write 3–4 sentences.

Key: Keep Quinn's line, $\hat{y} = 5.6 + 3.2x$, because it uses the correct response-SD over explanatory-SD ratio. Both lines pass through $(7, 28)$, but only Quinn's has the regression slope; passing through the mean point alone is not enough to minimize squared residuals. Keep Rowan's 64%: the linear model explains 64% of the variation in daily generation. Quinn's 80% confuses correlation with its square.